ICT5101 - Internet of Things
Part B - Coursework 2026

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Predictive Maintenance

The Specification

The dataset for this assignment is available at https://www.kaggle.com/datasets/canozensoy/industrial-iot-dataset-synthetic/data. It consists of synthetic data (around 500,000 records) of sensor readings of simulated machines deployed in a fictitious smart factory environment. Your task is to apply a Machine Learning algorithm (or algorithms) to predict maintenance needs based on usage.

You may use Weka, RapidMiner or code from scratch (e.g. using Python and supporting libraries). In the report accompanying the code, you should mention the tool used, assumptions made, justification of the ML model (i.e. what has influenced your choice and why is the algorithm suitable for the problem?) and an analysis of the results.

The Deliverables

The deliverables for the assignment are your report and a zip file containing the code. The main text of the report should not exceed 5 pages (i.e. The report content excluding Front page, table of contents and references). The guidelines can be found at https://www.acm.org/publications/authors/submissions, while the templates can be found at:

Word:
https://www.acm.org/binaries/content/assets/publications/taps/acm_submission_template.docx

LaTeX: https://www.overleaf.com/latex/templates/acm-journals-primary-article-template/cpkjqttwbshg

You may use up to an additional 10 pages as appendices and for referencing. You may not change font size, margins or line spacing. If you use Generative AI, you should add an appendix with the prompt used, justify why you used AI and the steps you took to ensure the correctness of the result.

Notes

- The deadline for the assignment is Friday 19th June, at midnight. It is worth 50% of your marks.
- You may, if you want to, use additional datasets to complement the one provided.
- Make sure that you define what you tried to achieve, justify your decisions and how you evaluated your solution.